



**SIMATS**  
ENGINEERING



**SIMATS**  
Saveetha Institute of Medical And Technical Sciences  
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

## DSA0601 Data Handling and Visualization

### CO2 – AT1 Chart Construction Test

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#### Question 1 – Healthcare Analytics

##### Dataset

| Patient | Age | BMI  | Blood Pressure | Disease Severity |
|---------|-----|------|----------------|------------------|
| P1      | 24  | 19.8 | 110            | Low              |
| P2      | 32  | 24.5 | 118            | Medium           |
| P3      | 41  | 28.1 | 132            | High             |
| P4      | 48  | 30.3 | 145            | High             |
| P5      | 56  | 33.2 | 152            | Critical         |
| P6      | 65  | 36.5 | 168            | Critical         |
| P7      | 38  | 26.2 | 124            | Medium           |
| P8      | 29  | 22.1 | 116            | Low              |

##### R Program

```
library(ggplot2)
```

```
health <- data.frame(
```

```
  Patient=c("P1","P2","P3","P4","P5","P6","P7","P8"),
```

```
  Age=c(24,32,41,48,56,65,38,29),
```

```
  BMI=c(19.8,24.5,28.1,30.3,33.2,36.5,26.2,22.1),
```

```

BloodPressure=c(110,118,132,145,152,168,124,116),

Severity=factor(c("Low","Medium","High","High",
"Critical","Critical","Medium","Low"),
levels=c("Low","Medium","High","Critical"))
)

ggplot(health,

aes(x=Age,y=BloodPressure,color=Severity,size=BMI))+

geom_point(alpha=0.8)+

scale_color_manual(values=c("green","yellow","orange","red"))+

labs(title="Healthcare Risk Analysis",

x="Age",

y="Blood Pressure",

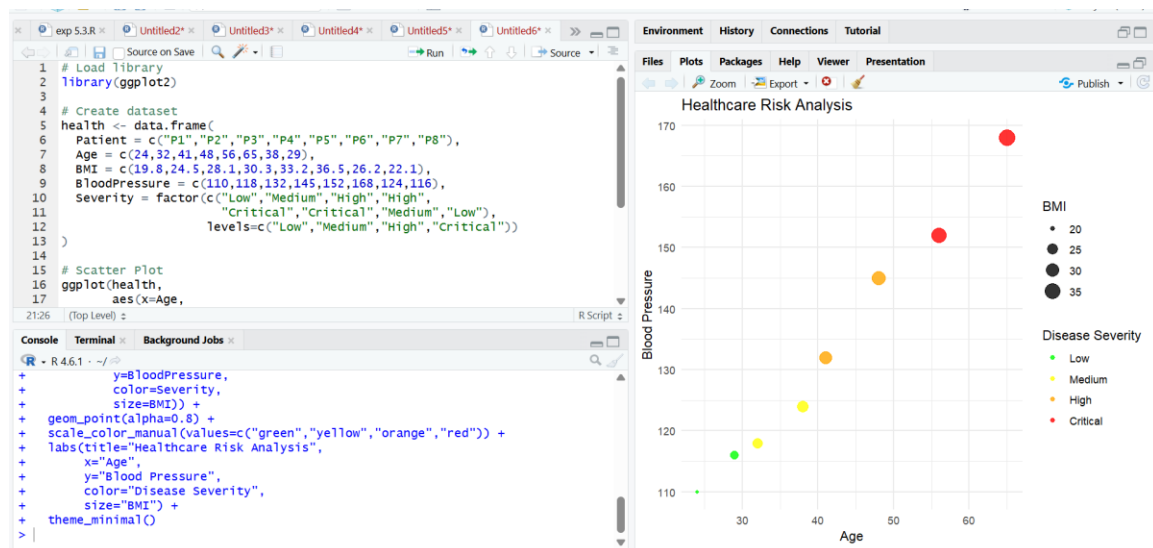
color="Disease Severity",

size="BMI")+

theme_minimal()

```

## OUTPUT:



## Justification:

- Age is placed on the X-axis because it is an independent variable.
- Blood Pressure is on the Y-axis to compare with age.
- Disease Severity uses a traffic-light color scale (green to red), making health risk easy to understand.
- BMI is represented by point size, where larger points indicate higher BMI.

## Question 2 – Climate Science

### Dataset

| Month | Solar | Temperature | Rainfall |
|-------|-------|-------------|----------|
| Jan   | 180   | 20          | 25       |
| Feb   | 200   | 23          | 18       |
| Mar   | 240   | 29          | 12       |
| Apr   | 280   | 35          | 8        |
| May   | 320   | 39          | 5        |
| Jun   | 300   | 36          | 65       |
| Jul   | 260   | 32          | 150      |
| Aug   | 250   | 31          | 170      |
| Sep   | 240   | 30          | 120      |
| Oct   | 220   | 28          | 85       |
| Nov   | 190   | 24          | 45       |
| Dec   | 170   | 21          | 30       |

### R Program

```
library(ggplot2)

climate <- data.frame(

  Month=c("Jan","Feb","Mar","Apr","May","Jun",
          "Jul","Aug","Sep","Oct","Nov","Dec"),

  Temperature=c(20,23,29,35,39,36,32,31,30,28,24,21),

  Rainfall=c(25,18,12,8,5,65,150,170,120,85,45,30)

)
```

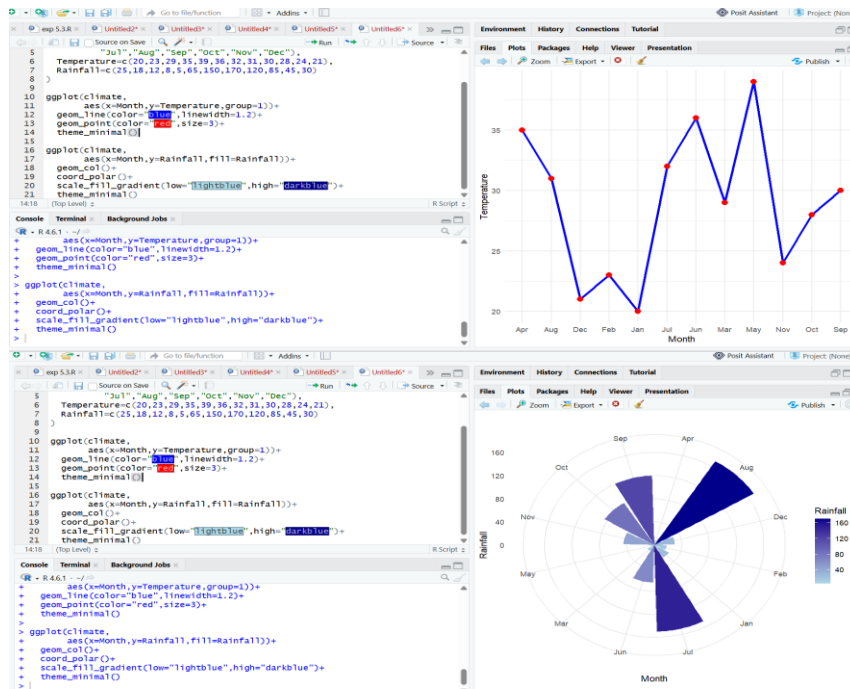
```

ggplot(climate,
aes(x=Month,y=Temperature,group=1))+
geom_line(color="blue",linewidth=1.2)+
geom_point(color="red",size=3)+
theme_minimal()

ggplot(climate,
aes(x=Month,y=Rainfall,fill=Rainfall))+
geom_col()+
coord_polar()+
scale_fill_gradient(low="lightblue",high="darkblue")+
theme_minimal()

```

## OUTPUT:



## Justification

### *Cartesian Chart*

- Clearly shows increasing and decreasing temperature.

- Easy to compare values.

### ***Polar Chart***

- Represents seasonal rainfall in a circular yearly pattern.
- Helps identify rainy and dry seasons.

## **Question 3 – Business Analytics and Marketing**

### **Dataset**

| Product | Sales | Profit | Market Share |
|---------|-------|--------|--------------|
| A       | 210   | 18     | 24           |
| B       | 185   | 14     | 20           |
| C       | 260   | 25     | 28           |
| D       | 145   | 12     | 15           |
| E       | 120   | 10     | 13           |

### **R Program**

```
library(ggplot2)

business <- data.frame(

  Product=c("A","B","C","D","E"),

  Sales=c(210,185,260,145,120),

  Profit=c(18,14,25,12,10),

  MarketShare=c(24,20,28,15,13)

)

business$Highlight <- ifelse(

  business$MarketShare==max(business$MarketShare),

  "Highest","Normal")

ggplot(business,

  aes(Product,Sales,fill=Profit))+

  geom_col()+

  geom_col(data=subset(business,
```

```

Highlight=="Highest"),

fill="red")+

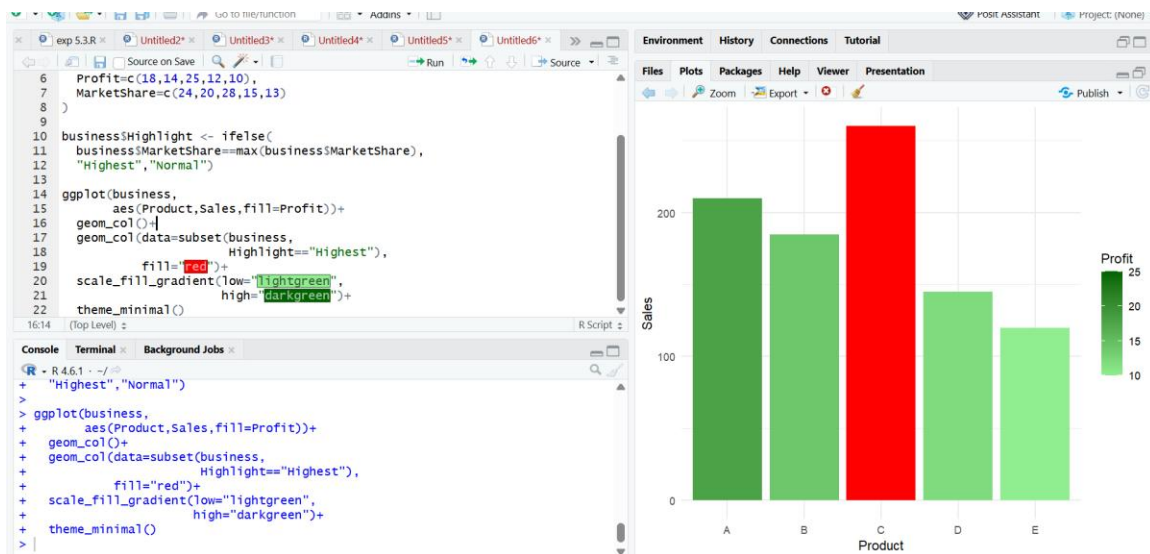
scale_fill_gradient(low="lightgreen",

high="darkgreen")+

theme_minimal()

```

## OUTPUT:



## Justification

- Bar height allows easy comparison of sales.
- Sequential green colors represent increasing profit.
- Red immediately highlights the highest market share product.

## Question 4 – Transportation Engineering

### Dataset

| Direction  | Vehicles/hour |
|------------|---------------|
| North      | 980           |
| North-East | 760           |
| East       | 1240          |

|            |      |
|------------|------|
| South-East | 890  |
| South      | 1420 |
| South-West | 1130 |
| West       | 860  |
| North-West | 720  |

## R Program

```
library(ggplot2)

traffic <- data.frame(

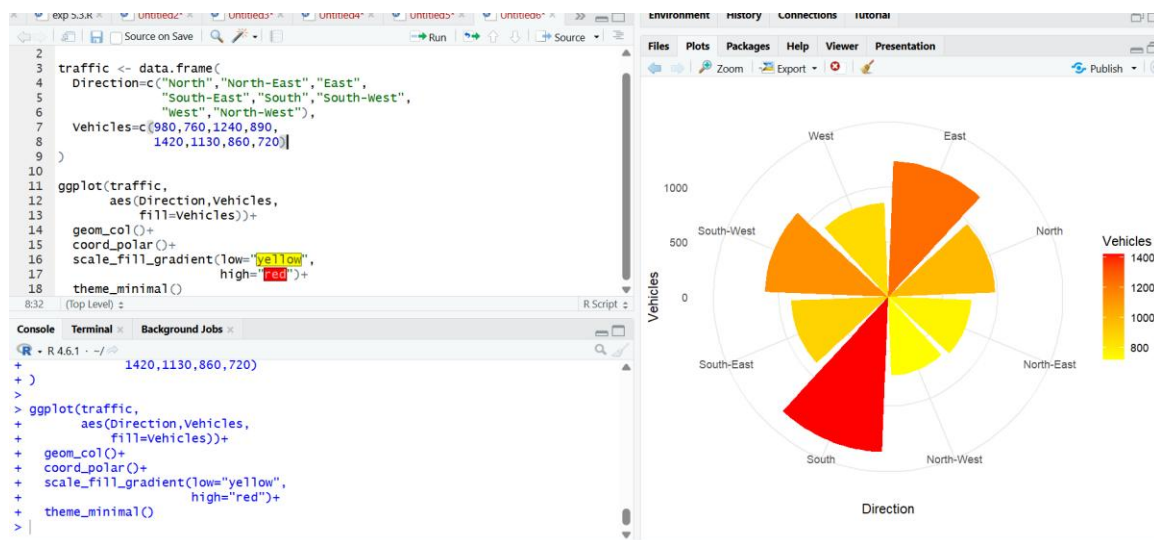
  Direction=c("North","North-East","East",
              "South-East","South","South-West",
              "West","North-West"),

  Vehicles=c(980,760,1240,890,
             1420,1130,860,720)

)

ggplot(traffic,
       aes(Direction,Vehicles,
           fill=Vehicles))+
  geom_col()+
  coord_polar()+
  scale_fill_gradient(low="yellow",
                     high="red")+
  theme_minimal()
```

## OUTPUT:



## Justification

- Polar chart naturally represents compass directions.
- Makes directional traffic comparison easier.
- Color gradient clearly indicates traffic density.

## Question 5 – Environmental Sustainability

### Dataset

| City | AQI | Green Cover | CO2 | Population |
|------|-----|-------------|-----|------------|
| A    | 48  | 45          | 1.8 | 9          |
| B    | 82  | 32          | 3.5 | 15         |
| C    | 135 | 21          | 5.6 | 22         |
| D    | 165 | 18          | 7.4 | 28         |
| E    | 72  | 38          | 2.9 | 12         |
| F    | 95  | 30          | 4.2 | 18         |

### R Program

```
library(ggplot2)
```

```
library(gridExtra)
```

```
city <- data.frame(
```

```
City=c("A","B","C","D","E","F"),
```

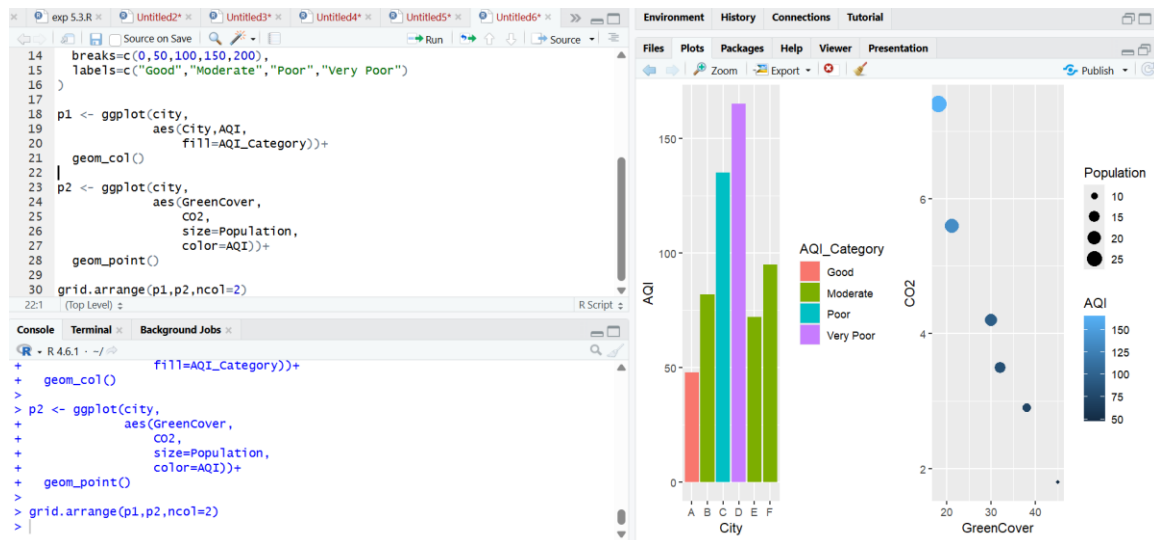


```

AQI=c(48,82,135,165,72,95),
GreenCover=c(45,32,21,18,38,30),
CO2=c(1.8,3.5,5.6,7.4,2.9,4.2),
Population=c(9,15,22,28,12,18)
)
city$AQI_Category <- cut(
city$AQI,
breaks=c(0,50,100,150,200),
labels=c("Good","Moderate","Poor","Very Poor")
)
p1 <- ggplot(city,
aes(City,AQI,
fill=AQI_Category))+
geom_col()
p2 <- ggplot(city,
aes(GreenCover,
CO2,
size=Population,
color=AQI))+
geom_point()
grid.arrange(p1,p2,ncol=2)

```

## OUTPUT:



## Justification

- Bar chart compares AQI among cities.
- Scatter plot shows the relationship between Green Cover and CO<sub>2</sub> emissions.
- Point size represents population.
- Green, Yellow, Orange, and Red colors effectively indicate AQI categories from good to poor air quality.